EDS 223 Homework Assignment 2

Exploring patterns of environmental justice

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Background

The Home Owners' Loan Corporation (HOLC) ranked houses by color, which led to the term "redlining". HOLC assigns grades from A to D, where A is the best and D is the worst grade. D is assigned the color red, terming "redlining". This is the basis of the social injustices to individuals living in these areas.

In this assignment, we explore the conditions of present-day injustices and the proportion of pedestrian and bicycle accidents by HOLC regions.

Data

```
library(tidyverse)
library(sf)
library(tmap)
library(here)
library(stars)
library(ggplot2)
library(patchwork)
```

```
# HOLC redlining
holcgrade <- st_read(here('data','mapping-inequality',
'mapping-inequality-los-angeles.json')) |>
  st_transform(crs = 4326)

# Environmental screening and mapping
ejscreen <- st_read(here('data','ejscreen',
'EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb')) |>
  st_transform(crs = 4326)
```

```r
# bike and pedestrian crash reports
crashes <- read_csv(here('data','SWITRS_LA','crashes.csv')) |>
  drop_na(POINT_X, POINT_Y) |>
  st_as_sf(coords = c('POINT_X','POINT_Y'), crs = 4326)
```

```r
# Check coordinate systems
st_crs(ejscreen) == st_crs(holcgrade) # TRUE
```

```
[1] TRUE
```

```r
st_crs(ejscreen) == st_crs(crashes) # TRUE
```

```
[1] TRUE
```

Part 1: Legacy of redlining in current environmental (in)justice

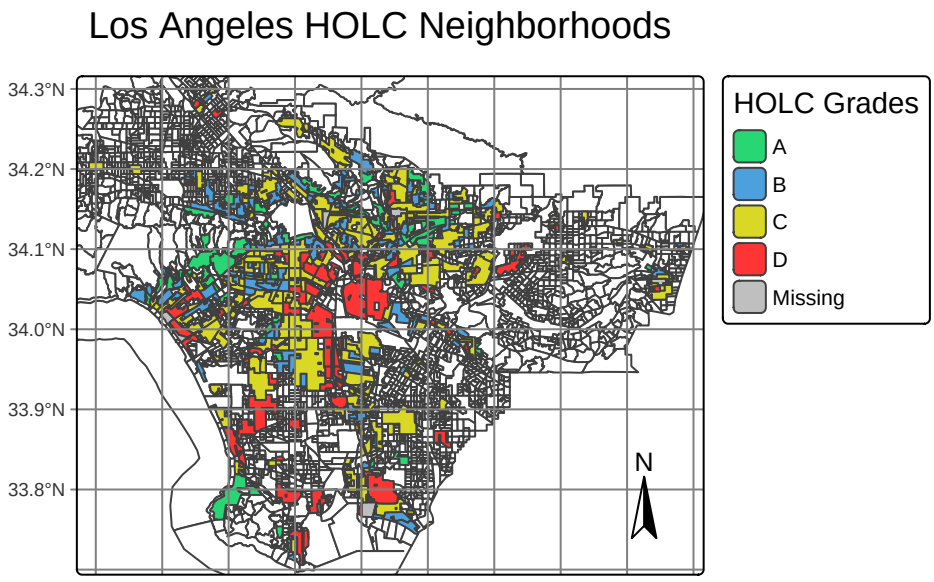
Exploring the historical redlining in Los Angeles and its legacy on present-day environmental justice.

1.1 Create a map of historical redlining neighbourhoods

```r
# Map of historical redlining neighbourhoods in LA
losangeles <- ejscreen |>
  filter(STATE_NAME == 'California', CNTY_NAME == 'Los Angeles County')

holc_colors <- c('#28d774','#4ca1dd','#d8d825','#ff3535')

tm_shape(losangeles, bbox = holcgrade) +
  tm_polygons(fill = 'white',
              lwd = 1) +
tm_shape(holcgrade) +
  tm_polygons(fill = 'grade',
              fill.scale = tm_scale_categorical(values = holc_colors),
              fill.legend = tm_legend(title = 'HOLC Grades',
                                      position = tm_pos_out("right"))) +
tm_title(text = 'Los Angeles HOLC Neighborhoods') +
tm_graticules() +
tm_compass()
```



This map shows the available information we have regarding HOLC grades. Now we know which areas we are evaluating. Visually, it seems like HOLC Grade C has the highest spatial coverage.

1.2.1 Summarize the data by percent of census block group by HOLC grade.

First, find the sptial overlap between the HOLC grade and Los Angeles.

```
# Join HOLC and LA spatially
holcgrade  <- st_make_valid(holcgrade)
losangeles <- st_make_valid(losangeles)

laholc <- st_join(losangeles, holcgrade, join = st_intersects) |>
  st_drop_geometry() |>
  select(ID, grade, LOWINCPCT, P_PM25, P_LIFEEXPPCT)
```

```
# tibble that shows the percent of census block groups in each HOLC grade
census_block_holc <- laholc |>
  group_by(grade) |>
  summarize(count = n()) |>
  mutate(perc_census = count/sum(count)*100)
census_block_holc
```

```
# A tibble: 5 x 3
  grade count perc_census
  <chr> <int>       <dbl>
1 A       449        5.00
2 B      1239       13.8
3 C      3058       34.0
4 D      1346       15.0
5 <NA>   2896       32.2
```

Based on the data, HOLC Grade C has the highest percent census block group per grade.

1.3 Create visualizations summarizing current conditions within HOLC grades

```
# finding the mean value per HOLC grade
conditions <- laholc |>
  group_by(grade) |>
  summarise(mean_low_pct = mean(LOWINCPCT, na.rm = TRUE),
            mean_pm25 = mean(P_PM25, na.rm = TRUE),
            mean_lifeexp = mean(P_LIFEEXPPCT, na.rm = TRUE))
conditions
```

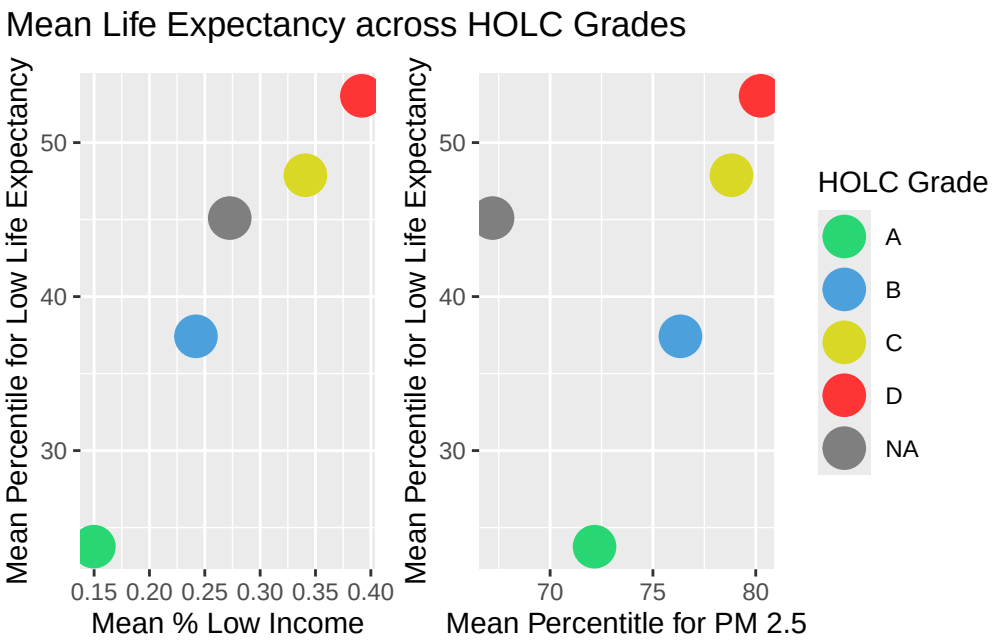
```
# A tibble: 5 x 4
  grade mean_low_pct mean_pm25 mean_lifeexp
  <chr>        <dbl>     <dbl>        <dbl>
1 A            0.150      72.2         23.8
2 B            0.242      76.3         37.4
3 C            0.341      78.8         47.9
4 D            0.392      80.2         53.0
5 <NA>         0.273      67.2         45.1
```

```
plot1 <- ggplot(data = conditions,
                mapping = aes(x = mean_low_pct,
                              y = mean_lifeexp,
                              color = grade)
) + geom_point(size = 7) +
  scale_color_manual(values = c('A'=holc_colors[1],
                                'B'=holc_colors[2],
                                'C'=holc_colors[3],
                                'D'=holc_colors[4])) +
  labs(x = 'Mean % Low Income',
```

```
       y = 'Mean Percentile for Low Life Expectancy',
       color = 'HOLC Grade') +
  theme(legend.position = 'none')

plot2 <- ggplot(data = conditions,
                mapping = aes(x = mean_pm25,
                              y = mean_lifeexp,
                              color = grade)
) + geom_point(size = 7) +
  scale_color_manual(values = c('A'=holc_colors[1],
                                'B'=holc_colors[2],
                                'C'=holc_colors[3],
                                'D'=holc_colors[4])) +
  labs(x = 'Mean Percentile for PM 2.5',
       y = 'Mean Percentile for Low Life Expectancy',
       color = 'HOLC Grade')

plot <- plot1 + plot2
plot + plot_annotation(title = 'Mean Life Expectancy across HOLC Grades')
```



1.4 Reflection

Based on the plot above, there is a strong pattern between HOLC grade and low life expectancy. As percent low income and PM 2.5 concentrations increase, low life expectancy increases as well. The grade with the highest values in all 3 categories is rank D, and vice versa for rank A.
This means that redlining grades tend to have low life expectancy, high percent low income, and high percentile for PM 2.5 as compared to A grade.

Part 2: Legacy of redlining in pedstrian and bike accidents

Explore the legacy of historical redlining in Los Angeles and pedestrian and bike crashes/fatalities.

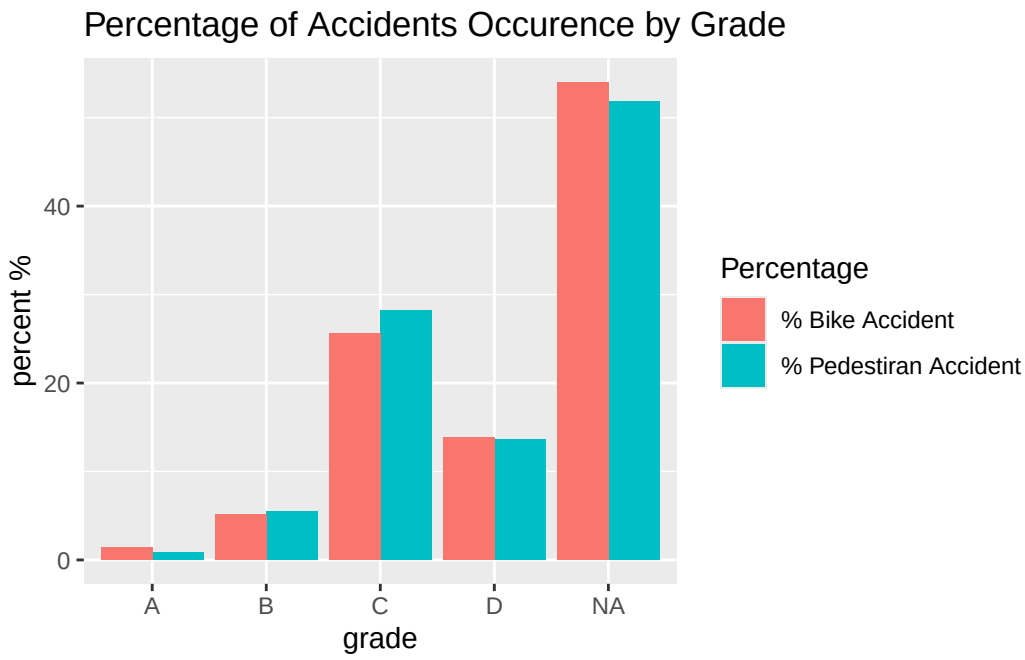
Create a figure that summarizes the percent of observations within redlined neighbourhoods within each HOLC grade.

```
# Finding the percent of observations for peds and bikes
crash_holc <- st_join(crashes, holcgrade, join = st_intersects) |>
  st_drop_geometry() |>
  select('grade','PEDESTRIAN_ACCIDENT','BICYCLE_ACCIDENT') |>
  group_by(grade) |>
  summarize(ped_accidents = sum(PEDESTRIAN_ACCIDENT == 'Y', na.rm = TRUE),
            bike_accidents = sum(BICYCLE_ACCIDENT == 'Y', na.rm = TRUE)) |>
  mutate(perc_ped_acc = ped_accidents/sum(ped_accidents) * 100,
         perc_bike_acc = bike_accidents/sum(bike_accidents) * 100)
crash_holc
```

```
# A tibble: 5 x 5
  grade ped_accidents bike_accidents perc_ped_acc perc_bike_acc
  <chr>         <int>          <int>        <dbl>         <dbl>
1 A               187            173        0.881          1.44
2 B              1171            612        5.51           5.10
3 C              5980           3077       28.2           25.6
4 D              2901           1660       13.7           13.8
5 <NA>          10997           6476       51.8           54.0
```

```
# Pivot longer to make plotting easier
crash_holc_longer <- crash_holc |>
  pivot_longer(cols = c(perc_ped_acc, perc_bike_acc),
    names_to = 'percentage',
    values_to = 'accident'
  )
```

```
ggplot(crash_holc_longer,
       aes(grade, accident, fill = percentage)
  ) + geom_col(position = 'dodge') +
    labs(
      title = 'Percentage of Accidents Occurence by Grade',
      fill = 'Percentage'
    ) +
    ylab('percent %') +
    scale_fill_discrete(labels = c('% Bike Accident','% Pedestiran Accident'))
```



Based on the table and histogram, the highest proportion of accidents are not in an identified HOLC grade, then followed by HOLC grade C. The second highest proportion of accidents occured in HOLC Grade D. This indicates most accidents occur in the lowest two HOLC grades.